Is Poisson-like spiking essential for general-purpose neural computation?

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Spike trains in the brain often look like a Poisson process. What is the role of Poisson-like spiking in neural computation?

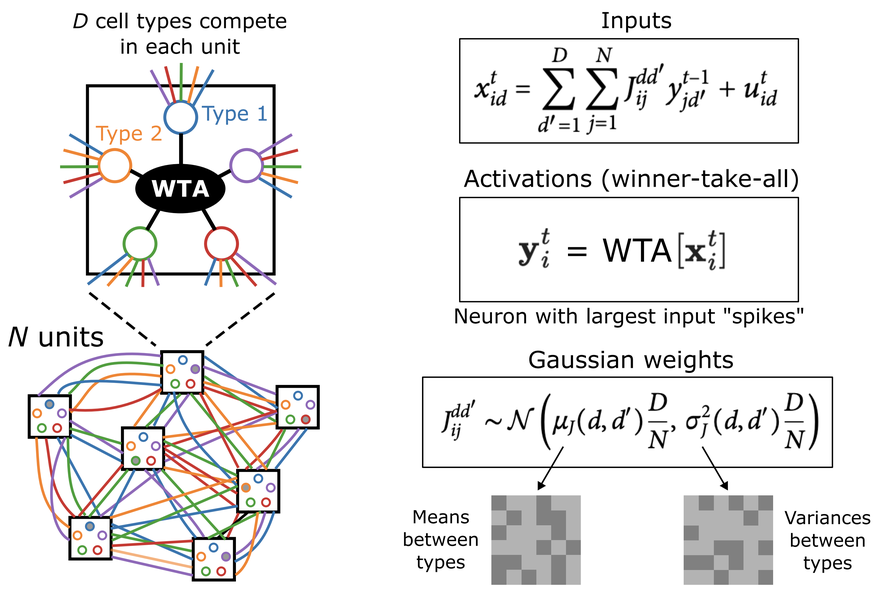
- Classic networks models produce Poisson-like irregular spiking but are limited in the types of computations they can perform [1].
- Despite the spiking nonlinearity, their population firing rate dynamics approach linearity in large networks [1,2].
- Ongoing efforts have sought to make irregular-spiking networks produce nonlinear dynamics, but approaches remain actively debated [2].
- Most models of computation through dynamics in the brain treat Poisson-spiking as observation noise [3].
- Here we show proof-of-concept for how Poisson-like irregular spiking can power nonlinear dynamics well suited for computation [4].

Main findings (Pang 2025. PLoS CB)

- A new spiking network model produces flexible nonlinear firing rate dynamics well posed for memory-dependent computation.
- These dynamics are powered by synaptic input fluctuations sustained by irregular spiking.
- The firing rate dynamics are well described by concise, differentiable mean-field equations, and can be trained or fit.

A new spiking network model based on winner-take-all competition among cell types

Network architecture and update equations



Inputs

$$x_{id}^t = \sum_{d'=1}^{D} \sum_{j=1}^{N} J_{ij}^{dd'} y_{jd'}^{t-1} + u_{id}^t$$

Activations (winner-take-all)

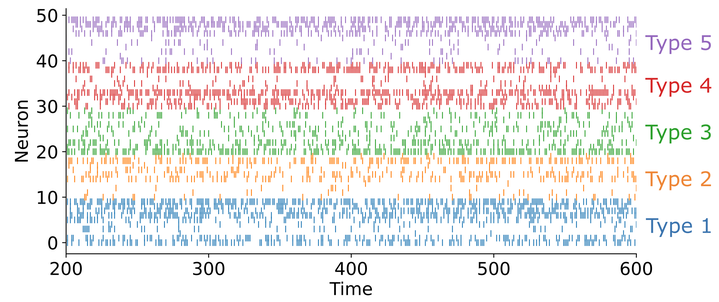
$$\mathbf{y}_i^t = \mathrm{WTA}[\mathbf{x}_i^t]$$

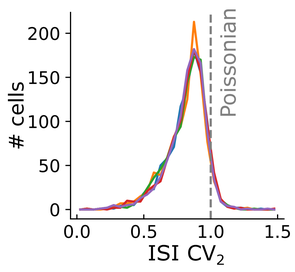
Neuron with largest input "spikes"

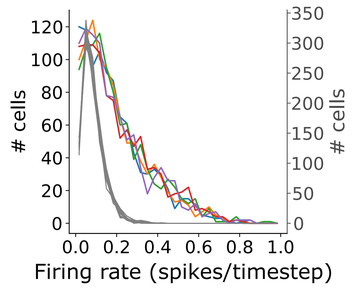
Gaussian weights

$$J_{ij}^{dd'} \sim \mathcal{N}\left(\mu_J(d,d')\frac{D}{N}, \sigma_J^2(d,d')\frac{D}{N}\right)$$

Emergent chaotic dynamics produce Poisson-like irregular spiking







Mean-field equations for firing rate dynamics

Synaptic input means

$$\mu_d \approx D \sum_{d'} \mu_J(d,d') r_{d'}^{t-1} + u_d^t$$

Synaptic input variances

$$\sigma_d^2 \approx D \sum_{d'} \sigma_J^2(d,d') r_{d'}^{t-1} + v_d^t$$

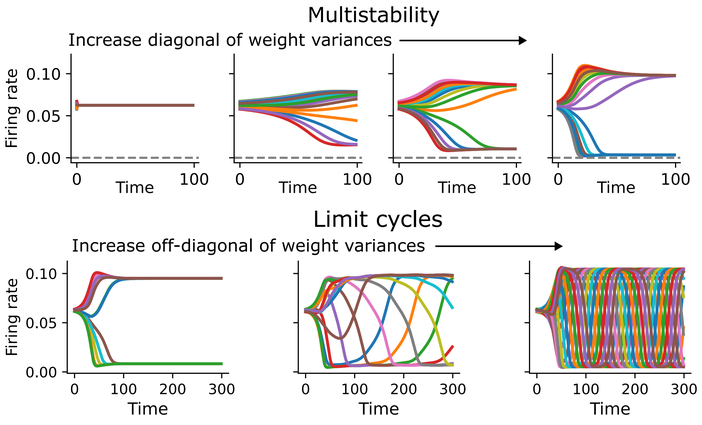
Firing rates

$$r_d^t \approx \int_x \mathcal{N}\left(x; \mu_d, \sigma_d^2\right) \prod_{d' \neq d} \Phi\left(x; \mu_{d'}, \sigma_{d'}^2\right) dx$$

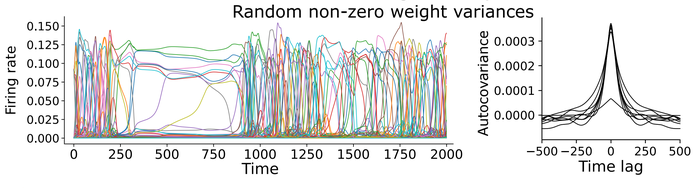
*differentiable

Theory predicts diverse, purely fluctuation-driven dynamics

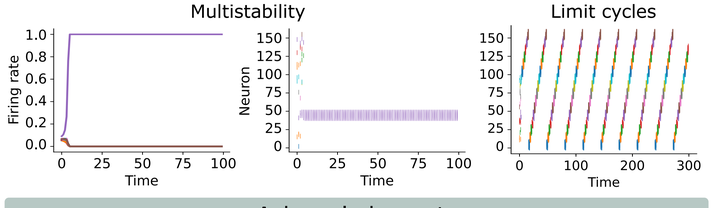
$\mu_J = 0 \Rightarrow \mu_d = 0$ Zero synaptic input means



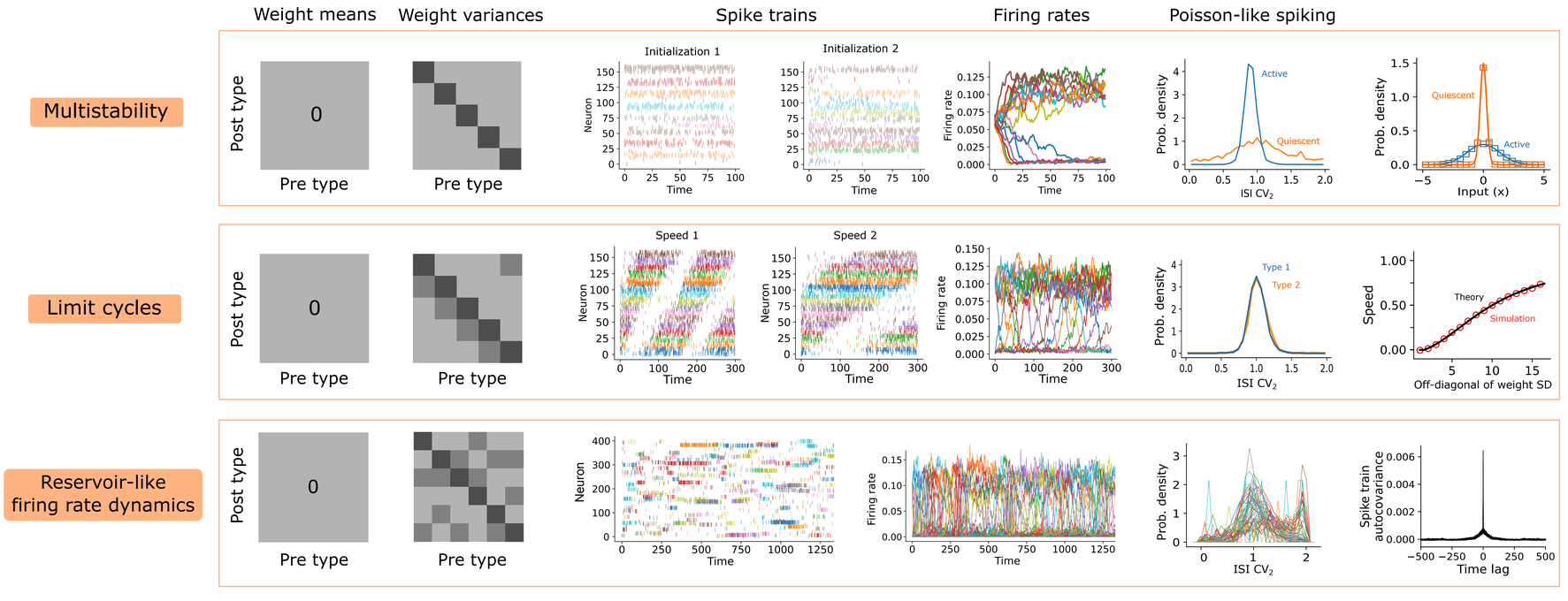
Reservoir-like firing rate dynamics



Mean-driven dynamics lead to saturated firing rates



Complex memory-dependent dynamics sustained by irregular spike trains emerge from editing only weight variances



Acknowledgments

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References

1. Van Vreeswijk & Sompolinsky 1996, 1998; Brunel 2001. 2. Thalmeier et al 2016; Gilra & Gerstner 2017; Alemi et al 2018; Ahmadian & Miller 2019; Baker et al 2020; Podlaski & Machens 2024. 3. Vyas et al 2020; Duncker and Sahani 2021; Pandarinath et al 2018. 4. Pang 2025.